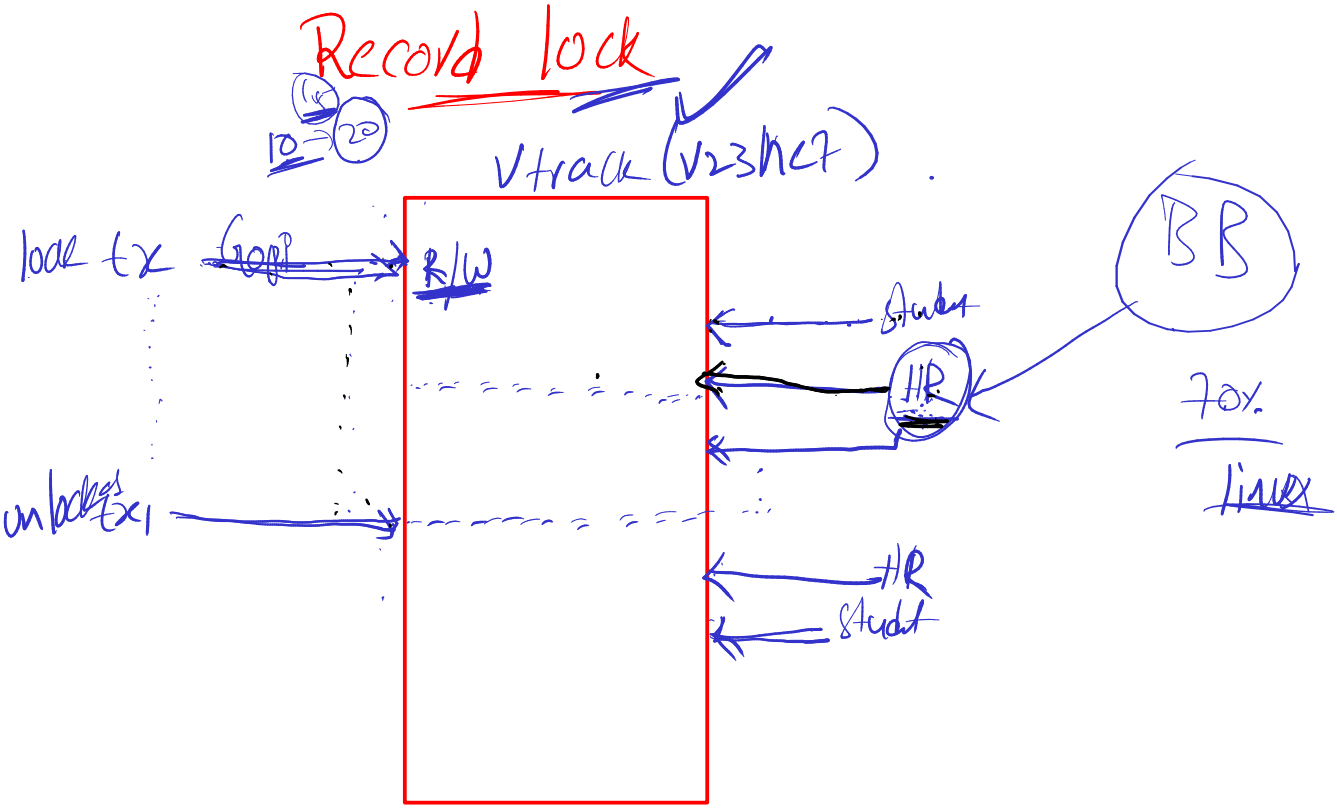
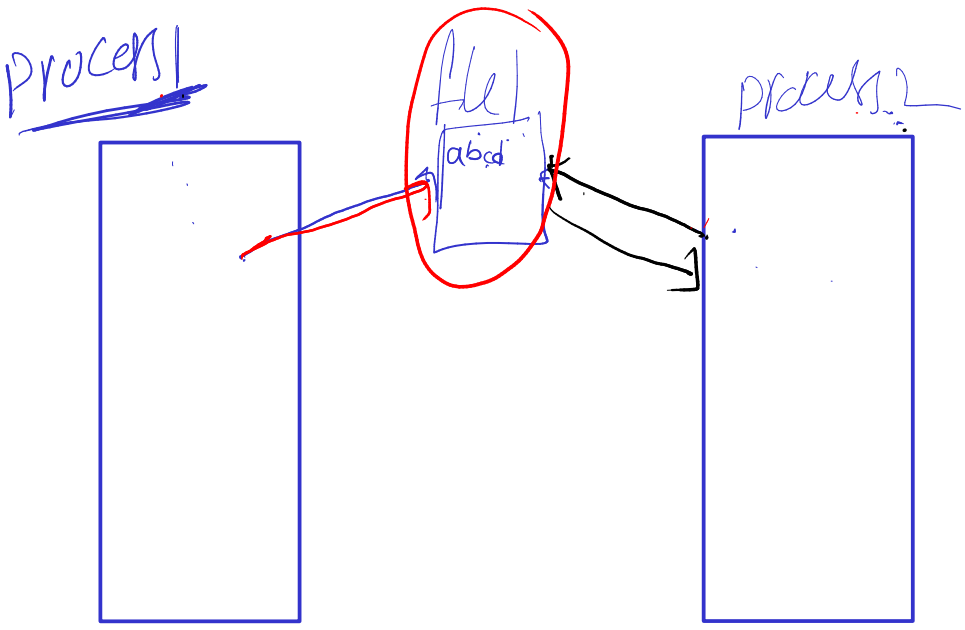

Record lock
10
20
Vtrack (V23K47)
lock tx
Group
R/W
unlock tx
BB
70%
HR
Studt
HR
Studt

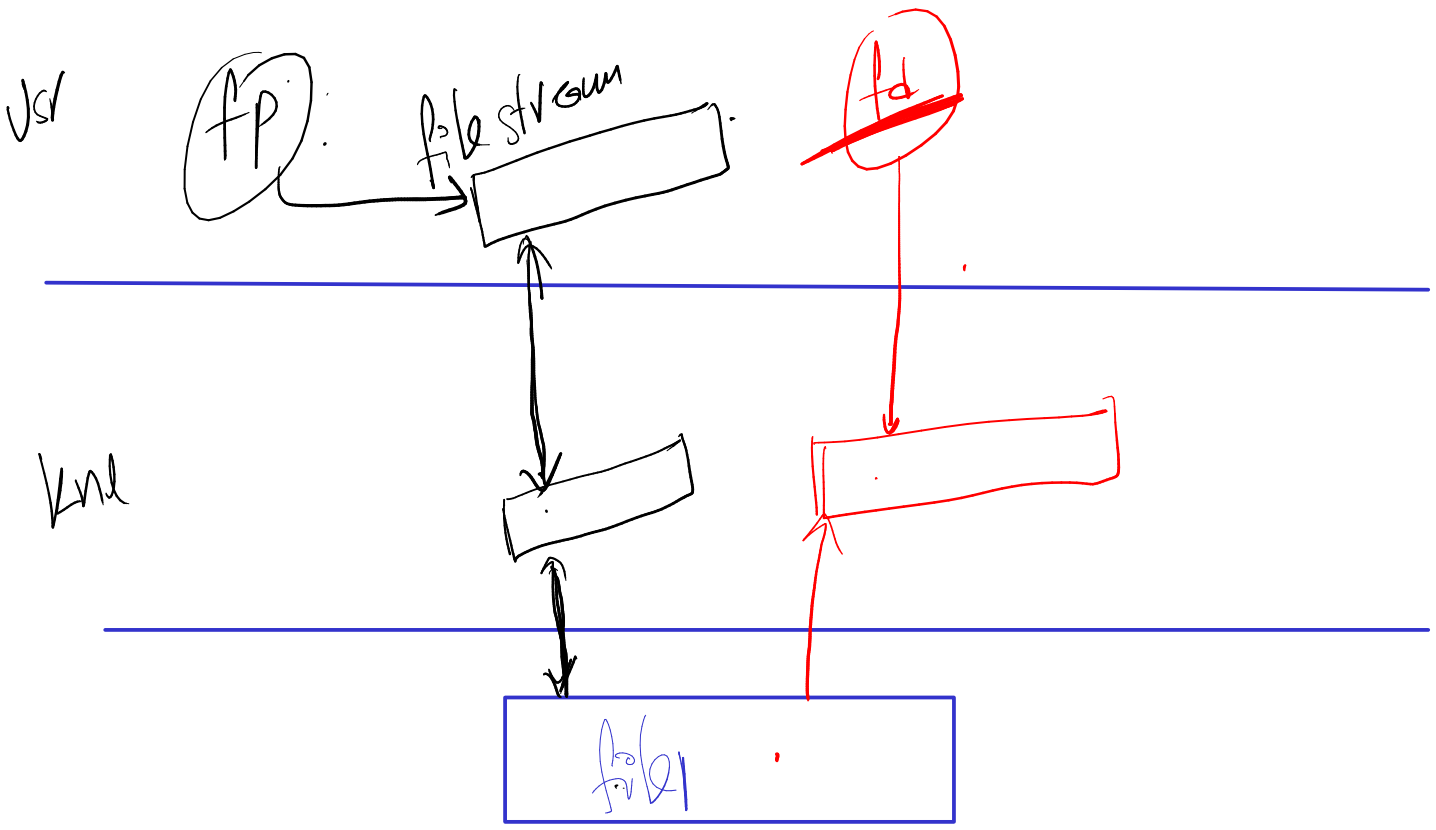

Process1
file
abcd
process2

fcntl

↳ file controlling

↳ Read lock, duplicate file des

↳ int



fcntl

```
int fcntl(int fd, int op, ... /* arg */ );
```

V.A.L

```
struct flock {
    ...
    short l_type;    /* Type of lock: F_RDLCK,
                        F_WRLCK, F_UNLCK */

    short l_whence;  /* How to interpret l_start:
                        SEEK_SET, SEEK_CUR, SEEK_END */

    off_t l_start;   /* Starting offset for lock */
    off_t l_len;     /* Number of bytes to lock */
    pid_t l_pid;     /* PID of process blocking our lock
                        (set by F_GETLK and F_OFD_GETLK) */
};
```

optional

if other job already applied lock, then it will wait untill unlock

F_SETLK, F_SETLKW, and F_GETLK

F_SETLK
- Test lock
- apply lock, if no one lock it
- if some other job already applied lock, then returns -1

F_GETLK
- Vi

prog1

int fd;

fd = open(____, O_WRONLY);

struct flock var;

var.l_type = F_WRLCK;

entire file locking // var.l_whence = var.l_start = var.l_len = 0;

→ fcntl(fd, F_SETLKW, &var);

```
char ch='a';
while(ch<='z')
{
    write(fd, &ch, 1);
    sleep(1);
    ch++;
}
close(fd);
```

unlock - fcntl